Affirm Holdings - Direction for Investment



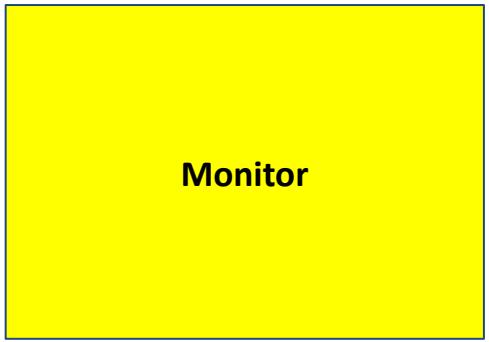
Executive Summary

IB is the profit engine, interest covers the credit risk. Split-pay and 0% are typically margin-negative



High Margin, low Loss

- Electronics (IB)
- Fitness (IB)
- Auto-Parts (IB)
- Home Furnishing (IB)
- Healthcare (IB)



Decent returns, moderate risk

- Fashion (IB)
- Travel (IB) - Conditional



MDR Heavy, margin negative

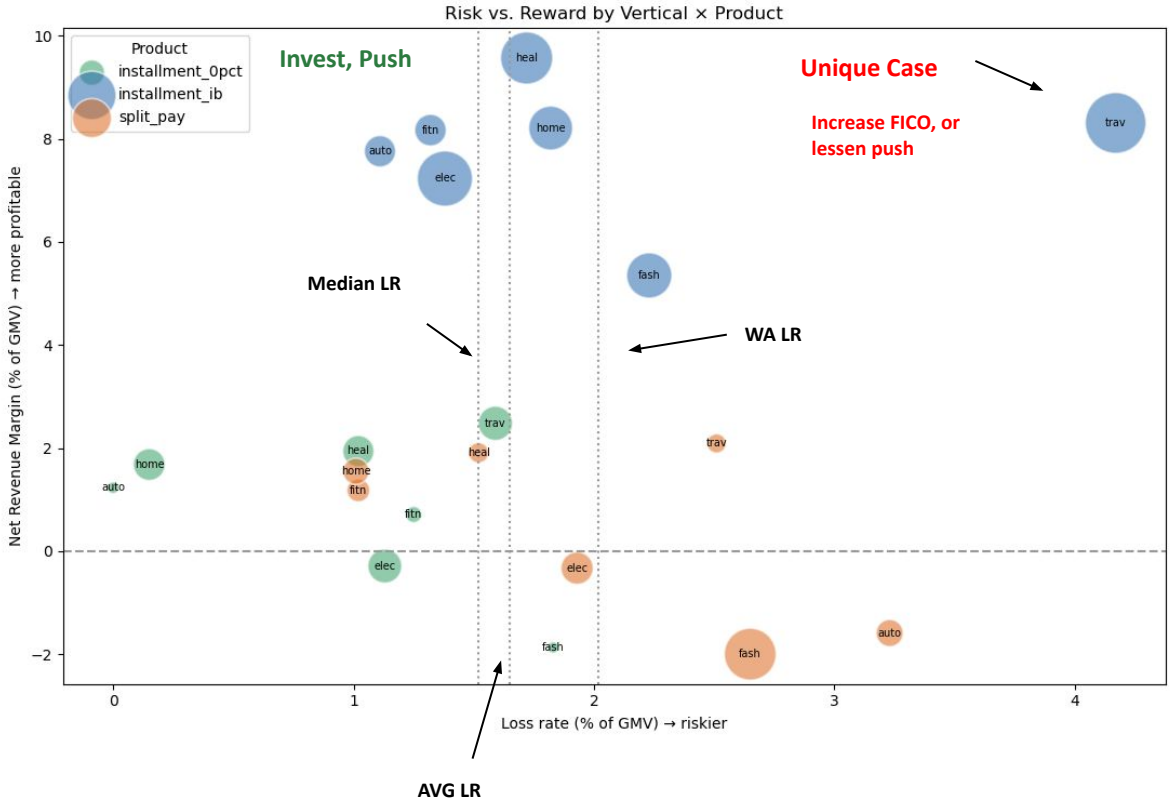
- Travel (Split Pay)
- Auto-Parts (Split Pay)
- Fashion (Split Pay)
- Electronics (IB 0%)

affirm

Product Mix is key decision -
Credit capacity is key in making Travel even more profitable, cutting losses on tight segment.

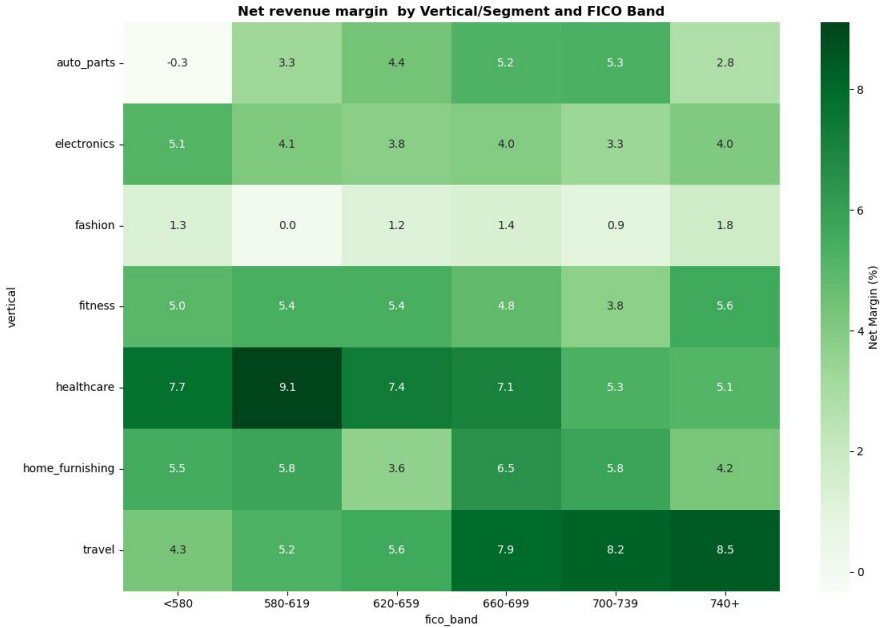
Interest-bearing drives margins; manage risk

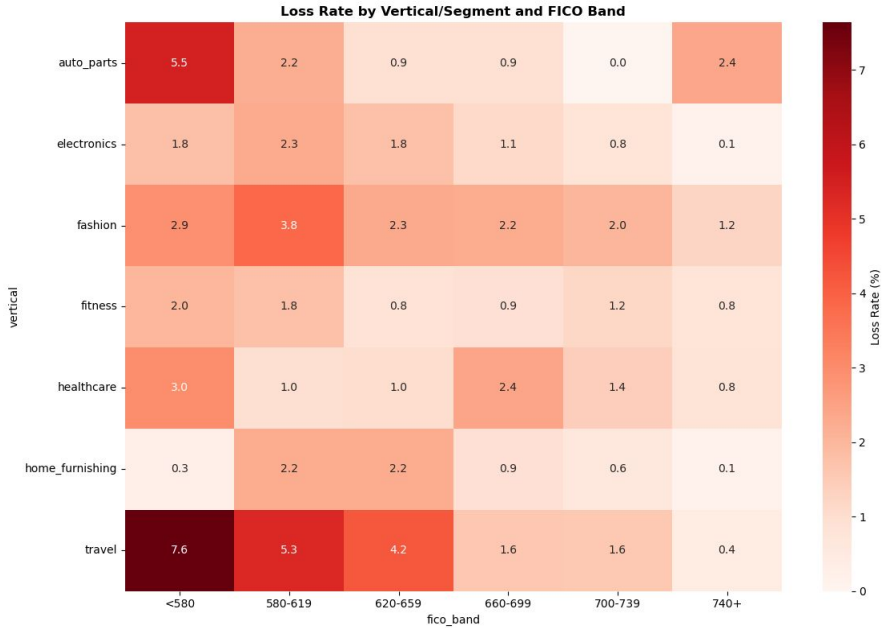
- **IB Segment** essentially clears the cost of risk by 5% - 8%.
- **Travel IB** is the riskiest bet, 4.17% LR rate, clearing cost of risk by 4%.
- Split Pay and 0% IB segments **hurt company**
    - No interest collected
    - Mostly Negative or tight net revenue $ and margins
    - Tougher to clear Risk



Risk lives in low-FICO tail

Travel loss lives at < 660 mostly, steadily decreasing over time. We see positive returns with higher FICO bands. **Exit lower FICO band, aim for above 660 FICO, for Travel to turn struggling segment to multiplied returns.**





affirm

Re-evaluate Travel IB

6.4 %

Travel IB LR rate on <660 FICO against 1.5 LR of > 660 FICO ~ 4.2x multiple, 485 bps spread

73%

of Travel IB losses from < 660

7% - 10 %

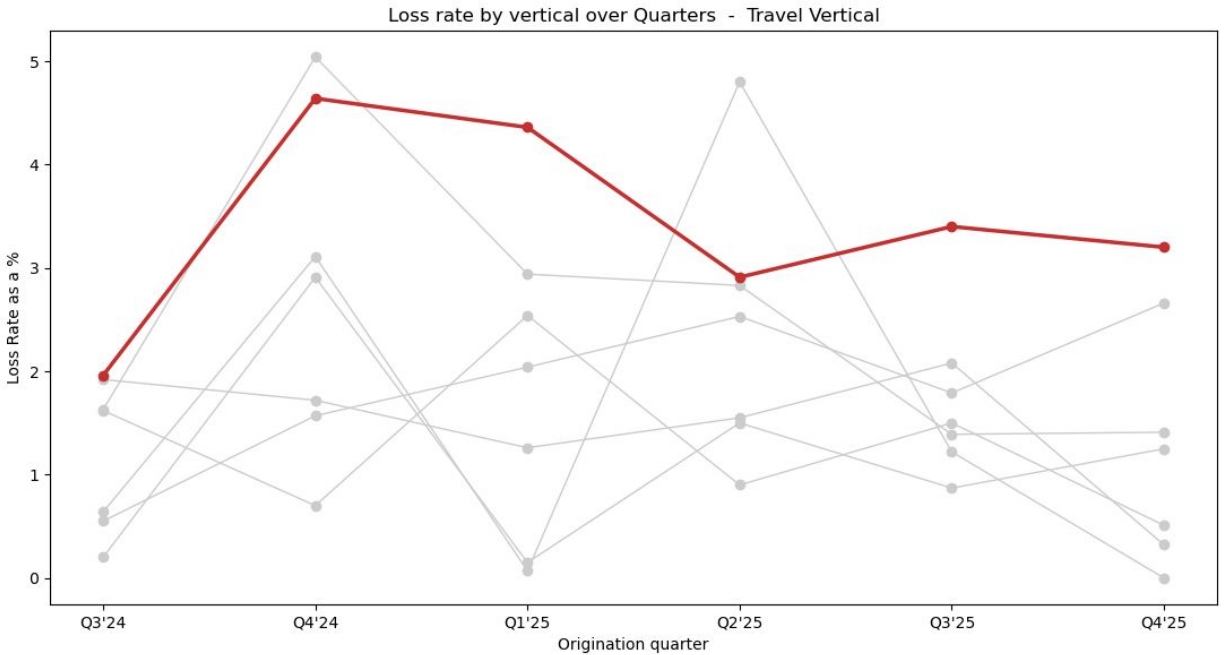
Redeployment Strategy into electronics, healthcare IB

Course of Action : **Raise** FICO floor for **Travel IB** band to ~ 660 or greater. This eliminates ~73% losses of Travel IB's credit losses, offering margin more breathing room. **Redistributing this credit capacity** into other segments, high returns, like electronics and healthcare IB is the best course of action if not raising floor for FICO.

affirm

Travel is not a one quarter instance

- Travel (Red), sits atop the highest loss rate amongst other segments in every quarter
- This is a pattern, consistently appearing QoQ
- Other verticals run amongst 3%, with **fashion** noticeable high in Q4’24, then decreasing



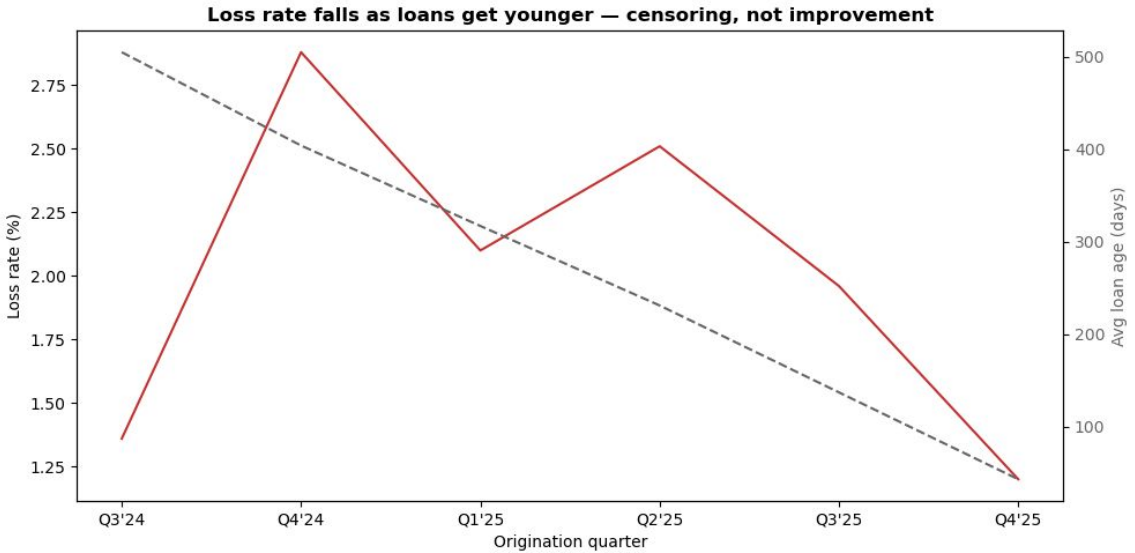
affirm

Appendix
Methodology - Data Integrity

affirm

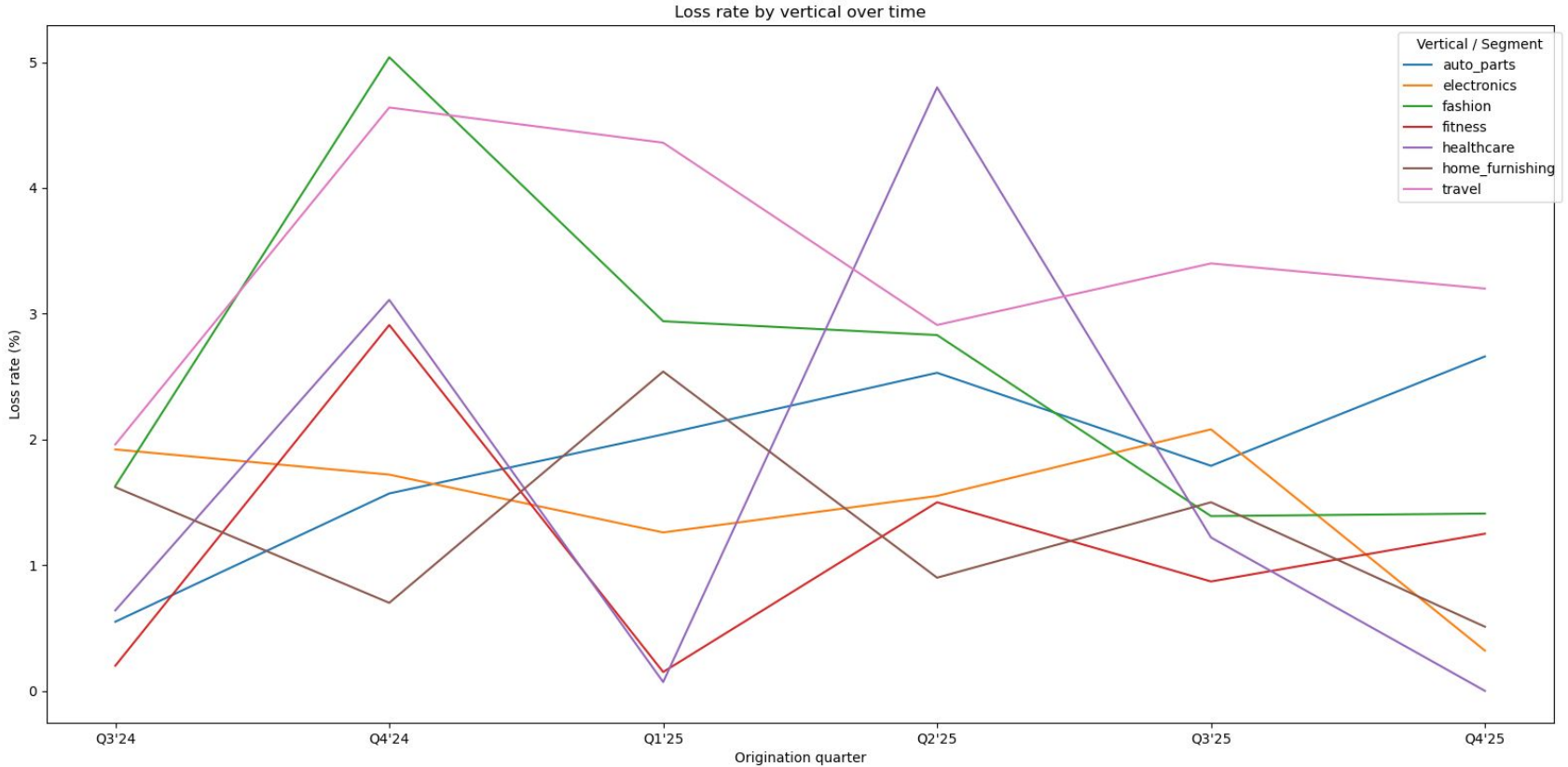
Why not trend raw loss ~ vintage bias

- Loans originated in late 2025 are too young to default
- Falling loss in recent quarters mirror loan age, so not true improvement
- Risk comparatively, seasoned vintages and recent quarter as floor for analysis



affirm

No vertical is consistently high except Travel



Insight on Loan Size amt

Travel IB has sizeable loan amount, proving there is consideration for new strategy plans, where we can redistribute into electronics, healthcare, fitness, etc.With **Fashion's Scale,** we can consider options to **partner with merchants** to **renegotiate MDR fees and percentages ~ segment strategy**



Break Down Financials

	vertical	product_type	gmv ($k)	net_revenue($k)	gross_revenue($k)	gross_cost($k)	interest collected ($k)	mdr fees ($k)	net_revenue_margin	loss_amount ($k)	cost to borrow ($k)	loss_rate	merchant_fee_as_a_%	merchant_fee_as_a_%_weighted	numbers_tie	clearing_margin	variance_of_margin
0	travel	installment_ib	547.63352	45.506242	87.00802	41.501778	43.17660	43.83142	8.309616	22.83530	18.666478	4.17	8.00	8.00	True	4.139616	0.992714
1	electronics	installment_ib	452.99964	32.748577	54.56428	21.815703	31.95051	22.61377	7.229272	6.24602	15.569683	1.38	4.98	4.99	True	5.849272	4.238603
2	healthcare	installment_ib	404.05499	38.668971	59.11332	20.444349	30.71817	28.39515	9.570225	6.93090	13.513449	1.72	7.02	7.03	True	7.850225	4.564084
3	fashion	split_pay	398.82987	-7.982014	16.04394	24.025954	0.00000	16.04394	-2.001358	10.55750	13.468454	2.65	4.02	4.02	True	-4.651358	-1.755230
4	fashion	installment_ib	304.41478	16.298650	33.39130	17.092650	21.14794	12.24336	5.354093	6.79461	10.298040	2.23	4.02	4.02	True	3.124093	1.400938
5	home_furnishing	installment_ib	287.42184	23.583237	38.46460	14.881362	21.16206	17.30254	8.205096	5.23125	9.650112	1.82	6.00	6.02	True	6.385096	3.508294
6	travel	installment_0pct	174.46370	4.320507	12.84023	8.519723	0.00000	12.84023	2.476450	2.77109	5.748633	1.59	7.39	7.36	True	0.886450	0.557516
7	electronics	installment_0pct	171.38224	-0.496489	7.29240	7.788888	0.00000	7.29240	-0.289697	1.93553	5.853359	1.13	4.24	4.26	True	-1.419697	-1.256369
8	electronics	split_pay	155.16849	-0.513670	7.75521	8.268880	0.00000	7.75521	-0.331040	2.99554	5.273340	1.93	4.98	5.00	True	-2.261040	-1.171524
9	home_furnishing	installment_0pct	150.39212	2.530398	7.93819	5.407792	0.00000	7.93819	1.682534	0.21927	5.188522	0.15	5.27	5.28	True	1.532534	10.216893
10	fitness	installment_ib	147.71874	12.066588	19.06947	7.002881	10.92764	8.14183	8.168624	1.95699	5.045891	1.32	5.51	5.51	True	6.848624	5.188352
11	auto_parts	installment_ib	146.47822	11.365585	17.88025	6.514665	10.52602	7.35423	7.759232	1.62807	4.886595	1.11	5.04	5.02	True	6.649232	5.990299
12	healthcare	installment_0pct	145.28066	2.817127	9.16680	6.349673	0.00000	9.16680	1.939093	1.48046	4.869213	1.02	6.34	6.31	True	0.919093	0.901072
13	auto_parts	split_pay	108.64971	-1.730500	5.42587	7.156370	0.00000	5.42587	-1.592733	3.50822	3.648150	3.23	4.99	4.99	True	-4.822733	-1.493106
14	home_furnishing	split_pay	101.65609	1.579295	6.03801	4.458715	0.00000	6.03801	1.553566	1.02692	3.431795	1.01	5.93	5.94	True	0.543566	0.538185
15	fitness	split_pay	78.14956	0.920250	4.34435	3.424100	0.00000	4.34435	1.177550	0.79930	2.624800	1.02	5.53	5.56	True	0.157550	0.154461
16	healthcare	split_pay	61.37978	1.171409	4.29444	3.123031	0.00000	4.29444	1.908460	0.93336	2.189671	1.52	7.05	7.00	True	0.388460	0.255566
17	travel	split_pay	54.93350	1.149482	4.44726	3.297778	0.00000	4.44726	2.092497	1.37962	1.918158	2.51	8.05	8.10	True	-0.417503	-0.166336
18	fitness	installment_0pct	39.17229	0.278352	2.03641	1.758058	0.00000	2.03641	0.710583	0.48839	1.269668	1.25	5.08	5.20	True	-0.539417	-0.431533
19	auto_parts	installment_0pct	20.05501	0.247263	0.88331	0.636047	0.00000	0.88331	1.232925	0.00000	0.636047	0.00	4.33	4.40	True	1.232925	inf
20	fashion	installment_0pct	19.25002	-0.359783	0.64429	1.004073	0.00000	0.64429	-1.869003	0.35160	0.652473	1.83	3.32	3.35	True	-3.699003	-2.021313

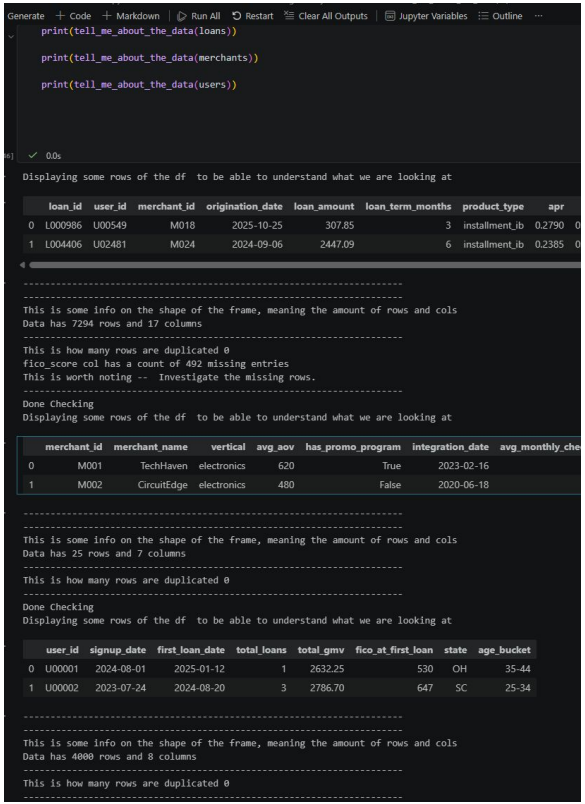
affirm

Charge-Off Accounts insight

vertical	product_type	loss_rate	chargeoff_gmv_%	chargeoff_cnt_%
travel	installment_ib	4.169814	10.460589	12.363636
auto_parts	split_pay	3.228927	3.479917	2.816901
fashion	split_pay	2.647119	2.686920	2.429728
travel	split_pay	2.511437	0.000000	0.000000
fashion	installment_ib	2.232024	2.344771	2.133713
electronics	split_pay	1.930508	2.270145	2.264151
fashion	installment_0pct	1.826492	2.481400	2.586207
home_furnishing	installment_ib	1.820060	1.083421	0.974026
healthcare	installment_ib	1.715336	2.064350	2.380952
travel	installment_0pct	1.588348	4.966380	4.901961
healthcare	split_pay	1.520631	0.000000	0.000000
electronics	installment_ib	1.378813	1.141844	1.355422
fitness	installment_ib	1.324808	0.890503	0.970874
fitness	installment_0pct	1.246774	0.000000	0.000000
electronics	installment_0pct	1.129364	1.319921	0.956938
auto_parts	installment_ib	1.111476	0.964157	1.162791
fitness	split_pay	1.022782	0.673030	1.111111
healthcare	installment_0pct	1.019034	0.000000	0.000000
home_furnishing	split_pay	1.010190	1.302972	1.470588
home_furnishing	installment_0pct	0.145799	0.000000	0.000000
auto_parts	installment_0pct	0.000000	0.000000	0.000000

affirm

Data Cleaning pt.1



```
Generate + Code + Markdown | Run All Restart Clear All Outputs | Jupyter Variables Outline ···
print(tell_me_about_the_data(loans))

print(tell_me_about_the_data(merchants))

print(tell_me_about_the_data(users))
```

0.0s

Displaying some rows of the df to be able to understand what we are looking at

	loan_id	user_id	merchant_id	origination_date	loan_amount	loan_term_months	product_type	apr	
0	L000986	U00549	M018	2025-10-25	307.85	3	installment_ib	0.2790	0
1	L004406	U02481	M024	2024-09-06	2447.09	6	installment_ib	0.2385	0

This is some info on the shape of the frame, meaning the amount of rows and cols
Data has 7294 rows and 17 columns

This is how many rows are duplicated 0
fico_score col has a count of 492 missing entries
This is worth noting -- Investigate the missing rows.

Done Checking
Displaying some rows of the df to be able to understand what we are looking at

	merchant_id	merchant_name	vertical	avg_aov	has_promo_program	integration_date	avg_monthly_che
0	M001	TechHaven	electronics	620	True	2023-02-16	
1	M002	CircuitEdge	electronics	480	False	2020-06-18	

This is some info on the shape of the frame, meaning the amount of rows and cols
Data has 25 rows and 7 columns

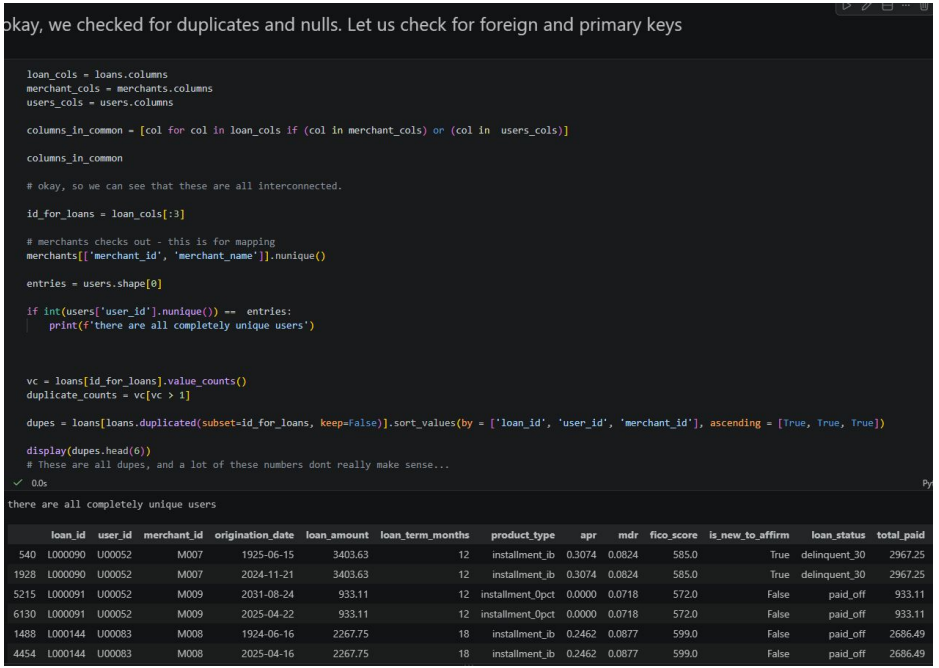
This is how many rows are duplicated 0

Done Checking
Displaying some rows of the df to be able to understand what we are looking at

	user_id	signup_date	first_loan_date	total_loans	total_gmv	fico_at_first_loan	state	age_bucket
0	U00001	2024-08-01	2025-01-12	1	2632.25	530	OH	35-44
1	U00002	2023-07-24	2024-08-20	3	2786.70	647	SC	25-34

This is some info on the shape of the frame, meaning the amount of rows and cols
Data has 4000 rows and 8 columns

This is how many rows are duplicated 0



okay, we checked for duplicates and nulls. Let us check for foreign and primary keys

```
loan_cols = loans.columns
merchant_cols = merchants.columns
users_cols = users.columns

columns_in_common = [col for col in loan_cols if (col in merchant_cols) or (col in  users_cols)]

columns_in_common

# okay, so we can see that these are all interconnected.

id_for_loans = loan_cols[:3]

# merchants checks out - this is for mapping
merchants[['merchant_id', 'merchant_name']].nunique()

entries = users.shape[0]

if int(users['user_id'].nunique()) ==  entries:
    print(f'there are all completely unique users')



vc = loans[id_for_loans].value_counts()
duplicate_counts = vc[vc > 1]

dupes = loans[loans.duplicated(subset=id_for_loans, keep=False)].sort_values(by = ['loan_id', 'user_id', 'merchant_id'], ascending = [True, True, True])

display(dupes.head(6))
# These are all dupes, and a lot of these numbers dont really make sense...
```

0.0s

there are all completely unique users

	loan_id	user_id	merchant_id	origination_date	loan_amount	loan_term_months	product_type	apr	mdr	fico_score	is_new_to_affirm	loan_status	total_paid
540	L000090	U00052	M007	1925-06-15	3403.63	12	installment_ib	0.3074	0.0824	585.0	True	delinquent_30	2967.25
1928	L000090	U00052	M007	2024-11-21	3403.63	12	installment_ib	0.3074	0.0824	585.0	True	delinquent_30	2967.25
5215	L000091	U00052	M009	2031-08-24	933.11	12	installment_0pct	0.0000	0.0718	572.0	False	paid_off	933.11
6130	L000091	U00052	M009	2025-04-22	933.11	12	installment_0pct	0.0000	0.0718	572.0	False	paid_off	933.11
1488	L000144	U00083	M008	1924-06-16	2267.75	18	installment_ib	0.2462	0.0877	599.0	False	paid_off	2686.49
4454	L000144	U00083	M008	2025-04-16	2267.75	18	installment_ib	0.2462	0.0877	599.0	False	paid_off	2686.49

affirm

Data Cleaning pt.2

okay, so the origination date is going to be the issue here.

It does not make sense they have loans from the future or past,

unless they inherited loans from the past, which is also unlikely.

we will filter based on affirms launch date and today's current data

```
july_start_date = pd.to_datetime('2024-07-01')

december = pd.to_datetime('2025-12-31')
# in theory, this should cut in half, so 160
dupes.shape[0]

mask = (dupes['origination_date'] >= july_start_date ) & (dupes['origination_date'] <= december)

# this worked !
# we can just apply a simple mask to the data to be able to retain the good data.
# we can bring all our dfs again
dupes[mask].shape[0]
```

22] ✓ 0.0s

affirm